

Practical program(18-09-2026)

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1. Virtual address Code:

```
sabiya@LAPTOP-DOV34MDC: × + v
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int x = 10;

    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}
```

OUTPUT SCREENSHOT :

```
sabiya@LAPTOP-DOV34MDC:~$ nano virtual.c
sabiya@LAPTOP-DOV34MDC:~$ gcc virtual.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Value of x = 10
Virtual address of x = 0x7ffddb07b054
sabiya@LAPTOP-DOV34MDC:~$ |
```

2. ProcessScape Code

```
GNU nano 8.7.1 processspace.c
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address : %p\n", (void *)main);
    printf("Data address : %p\n", (void *)&global);
    printf("BSS address : %p\n", (void *)&bss_variable);
    printf("Heap address : %p\n", (void *)heap);
    printf("Stack address : %p\n", (void *)&local);

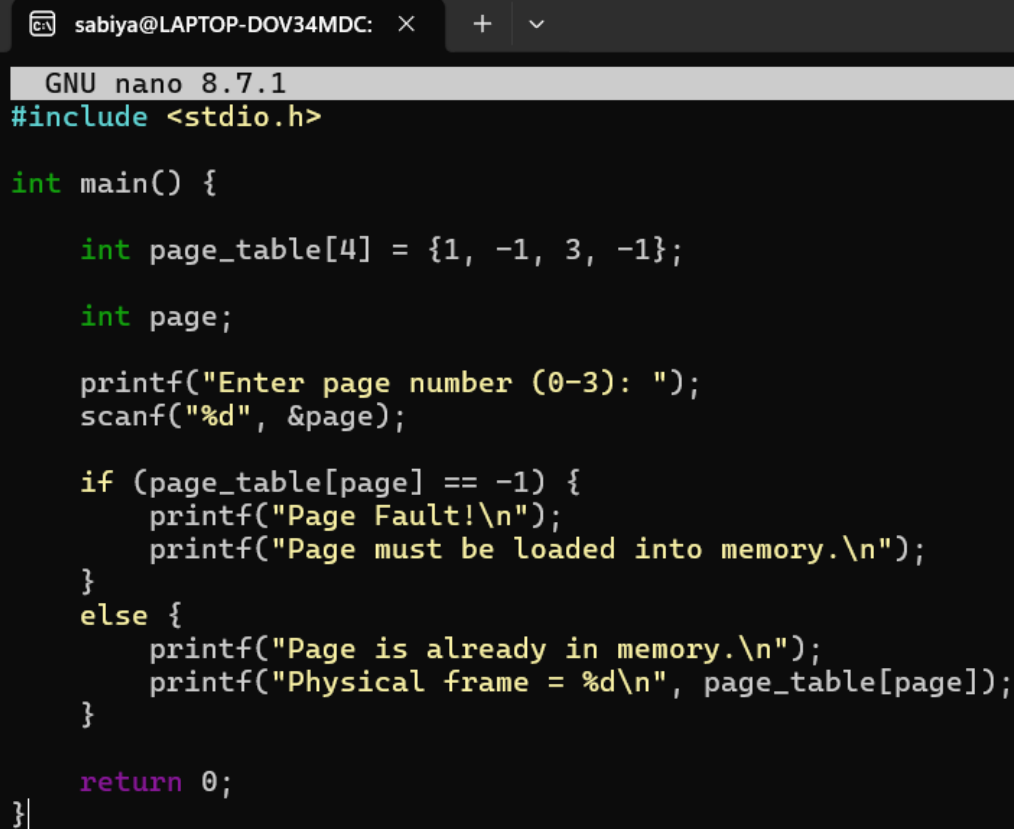
    free(heap);

    return 0;
}
```

OUTPUT SCREENSHOT:

```
sabiya@LAPTOP-DOV34MDC:~$ nano processspace.c
sabiya@LAPTOP-DOV34MDC:~$ gcc processspace.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Code address : 0x5d862961f1a9
Data address : 0x5d8629622010
BSS address : 0x5d8629622018
Heap address : 0x5d8655be0010
Stack address : 0x7fff3c28b4fc
sabiya@LAPTOP-DOV34MDC:~$ |
```

3.Demandpagging Code



```
GNU nano 8.7.1
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

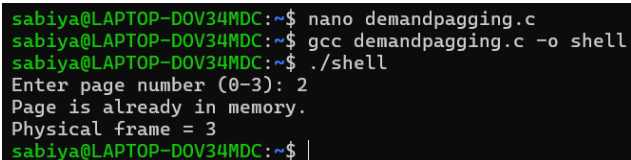
    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

OUTPUT SCREENSHOT:



```
sabiya@LAPTOP-DOV34MDC:~$ nano demandpagging.c
sabiya@LAPTOP-DOV34MDC:~$ gcc demandpagging.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
sabiya@LAPTOP-DOV34MDC:~$ |
```

4.Address Translation Code:

```
sabiya@LAPTOP-DOV34MDC: × + ▾
GNU nano 8.7.1
#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

    int physical_address = frame * page_size + offset;

    printf("Virtual Address = %d\n", virtual_address);
    printf("Page Number      = %d\n", page);
    printf("Offset           = %d\n", offset);
    printf("Frame Number      = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}
```

OUTPUTSCREEN SHOT:

```
sabiya@LAPTOP-DOV34MDC:~$ nano addressTranslation.c
sabiya@LAPTOP-DOV34MDC:~$ gcc addressTranslation.c
sabiya@LAPTOP-DOV34MDC:~$ gcc addressTranslation.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Virtual Address = 250
Page Number      = 2
Offset           = 50
Frame Number      = 1
Physical Address = 150
sabiya@LAPTOP-DOV34MDC:~$ |
```

5.malloc Code

```
sabiya@LAPTOP-DOV34MDC: × + ▾
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main() {

    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

    printf("\n");

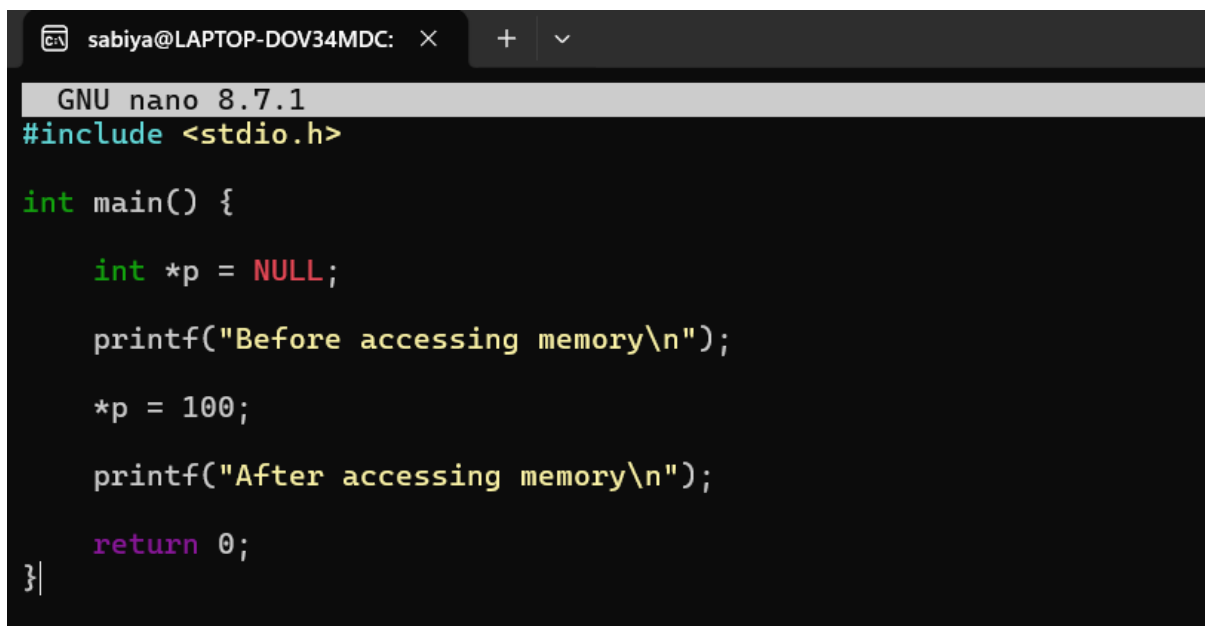
    free(p);

    return 0;
}|
```

OUTPUT SCREEN SHOT:

```
sabiya@LAPTOP-DOV34MDC:~$ nano malloc.c
sabiya@LAPTOP-DOV34MDC:~$ gcc malloc.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
10 20 30 40 50
sabiya@LAPTOP-DOV34MDC:~$ |
```

6. Segmentation Fault Code:



```
sabiya@LAPTOP-DOV34MDC: × + v
GNU nano 8.7.1
#include <stdio.h>

int main() {

    int *p = NULL;

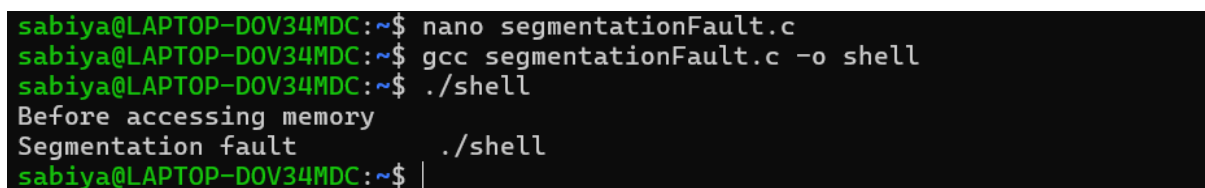
    printf("Before accessing memory\n");

    *p = 100;

    printf("After accessing memory\n");

    return 0;
}
```

OUTPUT SCREENSHOT:



```
sabiya@LAPTOP-DOV34MDC:~$ nano segmentationFault.c
sabiya@LAPTOP-DOV34MDC:~$ gcc segmentationFault.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Before accessing memory
Segmentation fault ./shell
sabiya@LAPTOP-DOV34MDC:~$
```